OMER BIN HABIB

Software Engineer · AI features · LLM Integration · Python · TypeScript · Node.js · PostgreSQL & pgvector

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SUMMARY

Software engineer with six years of professional experience, backend-focused, mainly Node.js, TypeScript and PostgreSQL. Built LLM and NLP features, including tool-use workflows with grounding constraints, schema-validated structured output, pgvector hybrid retrieval, and a FastAPI service built alongside Spot for PII redaction, outlier scoring, and sentiment gating. Treats model output as untrusted and designs systems to verify it before it affects application behaviour. I use AI coding agents daily under a written rules file.

EXPERIENCE

Independent Consultant · Remote · Karachi, Pakistan — Jan 2026 – Present

Backend engineer for UK SaaS clients, working asynchronously with distributed clients

- **Spot — UK B2B workplace wellbeing SaaS**. TypeScript and NestJS backend. Stripe Connect tiered billing, timezone-aware processing, and data handling built for GDPR from the start with hard delete, anonymisation and export flows. Alongside it I built a Python FastAPI NLP service: PII redaction with spaCy named-entity recognition and language detection, outlier scoring with an Isolation Forest, and sentiment gating with sentence embeddings. 357 automated tests and 18 database migrations maintained across the life of the project. Deployed on Render but never launched to users.
- **AgencyView — reporting platform for a UK SaaS founder**. PostgreSQL-driven engine producing thirteen report types with output filtered by user role. Ingestion runs as a scheduled pipeline with bounded concurrency, retry policies and structured logging, so failures are diagnosable without manual inspection.
- **Reliability by construction**: idempotent writes so a repeated job never duplicates data, retry with backoff on external calls, and work moved onto queues rather than held in the request path.
- **Remote and asynchronous by default** — requirements, trade-offs and release scope agreed in writing with clients in different timezones, with no shared office hours.

Software Engineer · Blocship · Karachi, Pakistan — May 2023 – Dec 2025

Backend development across nine products, in a team of Flutter and frontend developers plus QA; technically the only backend engineer

- **QR-code points and loyalty system for a vehicle-parts distributor**. NestJS backend handling point issuance, redemption and transfer between users — balance changes written transactionally so a failed transfer could not leave value in two places at once. I built the points reporting the distributor used to pay its buyers back according to the boxes they had bought.
- **MaidMe — home services marketplace for providers in Norway**. The Laravel backend was already live and had no support for socket connections, so rather than rewrite it I added a dedicated NestJS socket server alongside it — bringing real-time booking management and in-app support chat to a running product.
- **Disgo — streaming rooms with a social platform**. Built the Express and MongoDB backend on my own: room, feed and user APIs. It ran on a staging environment for a client demo.
- **Groops — social platform for organisations**. I did the backend as it moved from Lumen to Laravel to NestJS, covering group membership and engagement features, built on Clean Architecture: 16 modules, 73 migrations, 81 entities and 472 test suites.
- **Hubfiit and Apna Khata** — fitness and nutrition tracking; and an app for Pakistan's produce markets, where farmers sold fruit and vegetables to market traders — I built the transaction tracking between them. Laravel backends. Also Laravel work on Busy Jollof (a flow upgrade, plus invoicing and ticketing fixes), Street Delivery and Bizally (backend).
- **Across all nine**: REST API design documented with Swagger, JWT authentication and input validation, MySQL and PostgreSQL data modelling and migrations, code review, and production support — reproducing reported faults, tracing them through logs and shipping fixes under live conditions.

Software Engineer · Sybrid (Pvt) Ltd · Karachi, Pakistan — Apr 2021 – May 2023

Enterprise web applications for Japanese and Pakistani clients, in the Open Source team (Laravel, CodeIgniter, GrowCRM-type stacks) in the SDS department of a Lakson Group company

- **FiTE — IT engineer hiring portal for Japanese companies**. Laravel and Bootstrap, bilingual English/Japanese, role-based access across four user groups (candidate, agent, agency, client company) plus admin, security aligned to OWASP guidance. I owned

the candidate-hiring workflow from screening and interviews to onboarding, built with the stored procedures below, and built the admin panel (candidate form fields, form options, application-management settings). Released to AWS EC2 by building the release and uploading it as a zip; a DevOps colleague managed the instances.

- **MySQL stored procedures using window functions** to persist candidate profiles and calculate a profile score server-side, keeping scoring logic in one place rather than spread across request handlers.
- **Containerised a scheduled email service with Docker** — a reminder job for candidates who had begun but not completed an application, running behind nginx so the environment was reproducible rather than dependent on host configuration.
- **Multi-step profile form saved asynchronously per step**, so partial progress was never lost. Built with Bootstrap, jQuery and Ajax.
- **Colgate candidate hiring portal**. Job-post database structure and its process flow.
- **Colgate Inventory and Sales Management (support work)**. CodeIgniter system with a native mobile app and a web app; I fixed faulty data reports and built new ones.
- **Syvox (Sybrid's IP telephony and contact-centre platform)**. MySQL/PHP reports.
- **PMS (SybridMD claims and patient management, US market)**. Part of the development team on claims management.
- **cdprojectwing.com**. A customised GrowCRM (Laravel) for Canvas Digital, reshaped for project management; I used the repository pattern.
- **Small Node.js, Express and MongoDB CRUD tasks**.

Associate Developer · TAFSOL Technologies · Karachi, Pakistan Sep 2020 – Apr 2021

- **Athletes Guide — sports agency platform** (Laravel, Bootstrap). Registered coaches onboard new players through induction programmes, share academy training on AVA TV (a premium video library), and sell products through an online shop. I built the e-store and the rest of the platform, including the admin panel.
- **Timesheet management web app** for internal project management (Laravel, Blade, Bootstrap). It replaced the Excel sheets that production, QA and design staff used to log hours. I had started the next phase: reports on hours per team and per person.
- **Deployment**: deployed applications over FTP and cPanel (GoDaddy for domain and DNS, BlueHost for hosting).
- **Learned Laravel within a week** to get started.
- **Employee of the Month, October 2020**.

TECHNICAL SKILLS

Backend: Node.js, TypeScript, NestJS, Express, REST API design, JWT authentication, input validation, Swagger / OpenAPI, queue-based processing (BullMQ, Redis), WebSockets (Socket.io), scheduled and background jobs, idempotency, retry and backoff, transactional consistency, bounded concurrency

Databases: PostgreSQL and MySQL — data modelling, normalisation, indexing, views, migrations, stored procedures, window functions, multi-tenant schemas with row-level security

Other server-side languages: PHP (Laravel, Lumen), Python (FastAPI)

Cloud & DevOps: AWS — Lambda, RDS, S3, EC2; Docker, GitHub Actions, nginx, Render, Vercel, Supabase, Sentry, Git

Design & practice: Object-oriented design and dependency injection through NestJS, design patterns, code review, automated testing (Jest / Vitest, Mocha, Playwright end-to-end), ESLint

Frontend: React, Next.js (App Router), Redux, TypeScript, Tailwind

AI & LLM: Claude, OpenAI and Gemini APIs; tiered model routing — a cheaper model first, a stronger one only for ambiguous or high-stakes cases; schema-validated structured output (Zod); closed-set validation of model-supplied values; PII redaction before external calls; evaluation-gated model changes; prompt caching; pgvector with hybrid semantic and keyword retrieval; LLM tool-use loops with a grounding rule; LangGraph multi-agent workflows with Langfuse tracing; NLP with spaCy, scikit-learn and sentence-transformers

AI-assisted development: Claude Code as the main development method, under a written RULES.md the sessions are held to (reproduce before fixing, schema truth from migrations, no unverified numbers); project-scoped skills, sub-agents, slash commands, a PreToolUse hook and a tree-sitter code graph

Payments: Stripe Connect tiered billing

SELECTED PROJECTS

ALLDost · Activity and nutrition tracking web application Personal product

- **Next.js and React on a Supabase PostgreSQL backend** with row-level security, and an automated test suite maintained alongside it, including Playwright end-to-end tests.
- **Natural-language parsing of meals and workouts** into structured records using the Gemini API, with output validated against Zod schemas and values the model supplies checked against a fixed set rather than trusted.
- **Evaluation and review layer**: every AI-generated food entry is held as unverified until a reviewer approves or corrects it through a purpose-built review screen with its own access rules, covered by end-to-end tests.

Multi-agent build harness · LangGraph and Langfuse — Personal project

- **Orchestrator, specialist and critic agents that build software**, traced in Langfuse. Used it to build resume-tailor (public on GitHub), a resume generator that only includes claims backed by a verified profile.

EDUCATION

Master's in Business Analytics (MSBA) · Karachi School of Business & Leadership — 2025 – 2026

Final semester (In Progress)

BSc Computer Science · SZABIST · Karachi — Graduated 2020

CERTIFICATIONS

Introduction to Cloud Computing — IBM (2025) · Generative AI with Large Language Models — DeepLearning.AI with AWS · Microsoft Power BI · Building RESTful APIs in Laravel — LinkedIn Learning (2022)